

# Richard van Zyl — Skills Overview

## Solutions Architect / Technical Lead — Backend & Data-Intensive Systems

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**Years column.** Professional years only, counted from paid work starting **April 2017** — nothing exceeds **9**. Study years (2014–2016) are not counted.

### At a glance

| Area          | Strength                                                                        |
|---------------|---------------------------------------------------------------------------------|
| Primary stack | C# / .NET, SQL Server / Azure SQL, T-SQL                                        |
| Architecture  | Multi-tenant financial systems, DDD, CQRS, idempotent ledger design             |
| Cloud         | <b>Microsoft Azure</b> in production (not AWS)                                  |
| Delivery      | Azure DevOps YAML pipelines, Docker on Azure hosts, trunk-based + feature flags |
| Leadership    | ~5 years leading teams; ~3.5 as Solutions Architect / Technical Lead            |
| Analysis      | BRS / URS / FRS / TRS since 2017; stakeholder translation                       |

### Grouped skills (professional years)

#### Languages & data formats

| Skill                   | Years |
|-------------------------|-------|
| C# / .NET               | 9     |
| T-SQL                   | 9     |
| PostgreSQL              | 7     |
| TypeScript / JavaScript | 5     |
| JSON / XML              | 9 / 8 |
| YAML                    | 6     |
| Visual Basic .NET       | 3     |
| Java                    | 2     |

#### Databases & data engineering

| Skill                          | Years |
|--------------------------------|-------|
| SQL Server / SSMS              | 9     |
| SQL performance tuning         | 8     |
| Azure SQL                      | 7     |
| Entity Framework               | 9     |
| Redis (distributed caching)    | 5     |
| Table partitioning & archiving | 4     |
| In-Memory OLTP                 | 4     |
| MongoDB                        | 4     |
| SSRS                           | 5     |

## Architecture & design

| Skill                                 | Years |
|---------------------------------------|-------|
| Clean Architecture                    | 9     |
| Domain-Driven Design                  | 8     |
| Microservices                         | 7     |
| CQRS                                  | 6     |
| Event-driven architecture             | 5     |
| Multi-tenancy / tenant isolation      | 7     |
| Idempotency / exactly-once processing | 4     |

## Requirements & solution design

| Skill                                       | Years |
|---------------------------------------------|-------|
| Business & problem-domain analysis          | 9     |
| Requirements (BRS / URS / FRS / TRS)        | 9     |
| Solution design & architecture              | 9     |
| Technical documentation (audience-tailored) | 9     |
| Stakeholder engagement & translation        | 8     |
| Process & system modelling (UML)            | 8     |

## Backend frameworks & practices

| Skill                                   | Years     |
|-----------------------------------------|-----------|
| MediatR / AutoMapper / FluentValidation | 7 / 6 / 6 |
| OAuth2 / OpenID Connect / JWT           | 6         |
| Unit testing (xUnit / NUnit / Moq)      | 6         |
| Integration testing (TestContainers)    | 5         |
| Feature flags (Azure App Configuration) | 4         |
| Circuit breaker (Polly)                 | 3         |
| SignalR / WebSockets                    | 3         |
| Angular / Angular Material              | 6         |

## Cloud, DevOps & observability

| Skill                                         | Years |
|-----------------------------------------------|-------|
| Azure DevOps (previously TFS)                 | 9     |
| Azure CI/CD                                   | 7     |
| Pipeline-as-code (Azure DevOps YAML)          | 6     |
| Docker                                        | 7     |
| Message queues (Azure Service Bus / RabbitMQ) | 6     |
| Azure Monitor / App Insights / Log Analytics  | 5     |
| New Relic                                     | 4     |
| AKS / Kubernetes                              | 4     |
| API Gateway (Azure API Management)            | 4     |
| OpenTelemetry / Grafana / Graylog             | 3     |

Security & compliance

| Skill                                       | Years |
|---------------------------------------------|-------|
| Role-Based Access Control (RBAC)            | 8     |
| Data-protection-aware design (POPIA / GDPR) | 5     |
| SOX-aligned data segregation                | 4     |

Honest boundaries

Keep these framing notes when matching roles — they are deliberate, not soft-peddalling.

- **No AWS experience.** Cloud depth is **Microsoft Azure**. Do not map skills to “AWS equivalents.”
- **Infrastructure as Code:** declarative work is **Azure DevOps YAML pipelines** (pipeline-as-code) and **Docker on Azure hosts** — not Terraform, Bicep, or ARM templates in anger. Roles that lead on cloud resource provisioning as code need ramp-up; that is not a current strength.
- **Python:** not a language to hire for. Production exposure is **IronPython hosted in a C# backend** for integration marshalling (pre-v3), plus limited contemporaneous Python at MeterMo. Treat as read-and-modify.
- **Front-end:** not primary. Angular up to v8 and TypeScript are on the record, but front-end skills are not actively maintained; deep front-end leadership would need ramp-up.
- **Docker / Kubernetes:** comfortable with containerisation and high-level AKS (deployments, scaling); deep hands-on cluster ops is limited — would rely on operational support for that layer.
- **ActiveXperts:** used in production observability prose; not given a years row here (no invented span).

SQL performance tuning (condensed)

~8 years on high-volume financial ledger systems (150+ money processors across African markets).

| Focus                 | What it means in practice                                                          |
|-----------------------|------------------------------------------------------------------------------------|
| Execution plans       | SHOWPLAN, STATISTICS IO / TIME; spotting scans, lookups, high-cost operators       |
| Indexing              | Clustered / non-clustered / filtered / covering; fragmentation and rebuild choices |
| Locking & concurrency | Isolation levels; deadlock detection via Profiler / Extended Events                |
| Tuning patterns       | SARGable predicates; join vs EXISTS; avoiding scalar UDFs; subquery rewrites       |
| Scale design          | Date/tenant partitioning; indexed/materialised views; high-volume batch inserts    |
| In-Memory OLTP        | Memory-optimised tables and natively compiled procedures on hot paths              |
| Health monitoring     | Query Store, wait stats, index usage, sp_WhoIsActive                               |

Career timeline (condensed)

| Period              | Role                                                          | Organisation                                                                                                   |
|---------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Apr 2026 – present  | Co-owner / Solutions Architect                                | Yuno Technologies                                                                                              |
| Dec 2022 – Mar 2026 | Solutions Architect / Technical Lead — Core Financial Systems | Raging River Trading                                                                                           |
| Apr 2022 – Dec 2022 | Software Developer                                            | MeterMo (utilities metering; six projects TFS → Azure DevOps; APIs, field apps, websites; platform-debt close) |

| Period              | Role                                            | Organisation              |
|---------------------|-------------------------------------------------|---------------------------|
| Jan 2022 – Apr 2022 | Software Developer (Specialist Problem-Solver)  | Dotcom Software Solutions |
| Apr 2020 – Jan 2022 | Senior Developer / Business Analyst / Team Lead | Payteq                    |
| Oct 2017 – Apr 2020 | Software Developer / Business Analyst           | iPlan Global              |
| Apr 2017 – Oct 2017 | Software Developer (C# .NET)                    | Novigo                    |

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## Education & assessment

- **National Diploma — IT: Software Development**, Varsity College, 2014–2016
  - **IKM C# Assessment**: 73rd percentile (assessed twice, six months apart)
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